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Towards predictive maintenance: the case of the aeronautical industry

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Abstract

Periodic maintenance of aeronautical equipment is an expensive process. However, the most practiced types of maintenance in the field of aeronautics are preventive and corrective maintenance. The aeronautical wiring companies produce a huge amount of data that, with proper processing and intelligent systems, can provide useful information and data for the reduction of machine downtime, therefore the implementation of predictive maintenance whose objective is to predict the failure of systems by continuously observing their status, in order to plan and carry out maintenance actions in advance. In this paper, the benefits of artificial intelligence are presented with machine learning and data mining techniques could be used by aeronautical wiring companies, which use preventive and corrective maintenance. In addition, a study of the current situation is made, as well as a proposal of an expert system based on machine learning for the development of predictive maintenance.

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1. Introduction

The aircraft as a whole is a complex system that contains miles of electrical wire harnesses that interconnect various sub-systems and components located throughout the aircraft. Most of the miles of wire found in use on aircraft today perform a critical function in the aircraft. Aviation, like any other industry, relies heavily on quality assurance to provide the best possible experience for their customers, whether for luxury or business, therefore disappointment to meet the standard regulations and compliance direction within the aerospace industry can lead to serious consequences.

Aeronautical wiring companies produce wire harnesses that must meet their customers' demands on time. However, many constraints have been reported including unplanned machine downtime that can impact delivery time.

In these companies of aeronautical wiring, we find the preventive maintenance and the corrective maintenance, the use of these two types of maintenance is a costly process and does not allow any more to answer the current aeronautical requirement, so we can point out that the unplanned stops of machines generate a lot of delays of the production, in other words a huge impact on the delivery times.

The aeronautical wiring companies produce a huge amount of data, which are related to the breakdowns and the maintenance interventions carried out on the machine park whether it is in preventive or corrective, among these data we find the details of each breakdown as well as the operations carried out for the recovery of the machine and the spare parts. This amount of data can be used to predict failures through predictive maintenance.

In this work we propose a maintenance management model in BPMN model that optimizes and makes reliable the maintenance intervention on wire cutting machines as well as electrical test equipment, based on an expert system that allows to drastically reduce the maintenance intervention time, and consequently the reduction of the machine downtime.

The structure of this paper is organized as follows: a theoretical background of maintenance and its types is provided in section 2. In Section 3, we provide an overview of the existing situation, and in the last section, we conclude the main contributions and the works future.

2. Types of maintenances

Maintenance is a business process capable of attributing profitability, productivity and safety to any business. Maintenance is defined as the company's process for keeping equipment in a desirable state of efficient operation and aims at increasing equipment reliability, improving production time and reducing costs [1].

Usually, the maintenance management is categorized into different policies: (i) corrective or run-to-failure maintenance, (ii) preventive maintenance, and (iii) predictive maintenance [2].

Corrective maintenance is reactionary and can occur on an unscheduled basis. It is the repair or replacement of an asset after a defected fault has taken place. It is generally carried out after the failure of an aircraft asset, and the objective is to restore it to a functioning state as soon as possible, either by repair or replacement [3]. Preventive maintenance on the other hand occurs on a scheduled basis [4] and is intended to reduce the probability of asset failure or degradation. Preventive maintenance in this sense requires assets to have a defined expected life span or measurable degradation [5] at which they are required to be replaced accordingly.

The basic principle of the Preventive Maintenance is the execution of maintenance tasks according to the manufacturer's standards. It is held in limited by the manufacturer deadlines and spare parts are replaced or equipment components that have been determined by the operating manual in order to minimize the malfunctions and preserve the right application of the equipment [1]. This type of maintenance is performed while the equipment is still working and is planned so that any required resources are available. So, predictive maintenance is a philosophy or attitude that uses the actual operating condition of the plant equipment and systems to optimize the plant operations and/or processes [6]. Predictive maintenance concerns the application of sensor technology and analytical tools to predict when equipment' failures might occur and to prevent the occurrence of the failures by performing maintenance. The failures' prediction can be done by applying, e.g., vibration monitoring or thermography, and must be effective at

predicting failures and also provide sufficient warning time for the upcoming maintenance. When this policy is working effectively, maintenance is only performed on equipment when it is required. [2]

Although predictive maintenance is not a new concept, different levels of maturity have been introduced. The easiest level is through a periodic visual inspection and utilizing a technician's experience and checklist to discover the best time to repair or change a piece of equipment. In the next level, periodic inspections are based on the technician's skill and using an instrument to read or record the status of the equipment. For performance measurement, the trained inspectors predict the problem by analyzing the data. In the third level, real-time condition monitoring is done by sensors that uninterruptedly collect and send data based on defined rules to know at what time maintenance should be done. [7,8].

Fig1 summarizes the three basic types of maintenance. As can be seen, the corrective maintenance take place after the failure occurrence; on the other hand, there are also preventive maintenance activities that can be applied in the pre- and post-failure periods; finally, predictive maintenance deals with errors in advance [9,10].

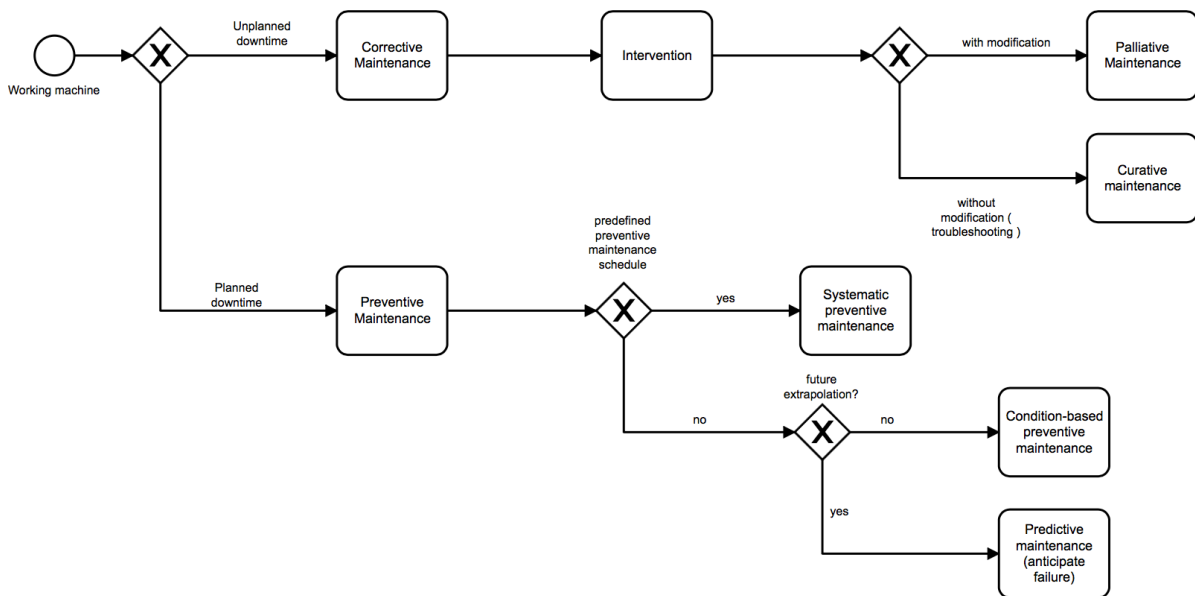


Fig. 1. Types of maintenance.

3. Overview of the existing situation

The maintenance practiced in aeronautic industries is the preventive and corrective maintenance.

Indeed, unplanned downtime is frequent and has a negative impact on production and disrupts production lines, causing unplanned delays that can impact the delivery time and can generate customer complaints.

Maintenance as it is practiced is represented by BPMN (Business Process Model and Notation) as shown in Fig.2, which defines the sequences of maintenance tasks performed within the shopfloor. By using BPMN notation, it is possible to represent a process, validating it and guaranteeing its consistency, with the same meaning as the textual/documentary description of the existing organizational processes [10].

In general, there are frequent unplanned downtimes of the wire Cutting machines and electrical testing equipment, time-consuming interventions, unreliable maintenance actions as well as frequent breakdowns.

In effect, following a breakdown, maintenance is called to intervene on the machine, and to assign a technician it takes a lot of time to understand the type of failure is it a mechanical, electrical, pneumatic or information technology and decide thereafter on the resource that must be assigned according to the machine failure, then we observe a huge loss of time that is related to the assignment of the technician before starting the intervention. In addition, the diagnosis takes a lot of time to identify the causes of the breakdown because of the lack of a reference system that capitalizes on the failure modes for each breakdown and thus optimizes the search for the causes, as the data concerning the failure is not used to capitalize and establish standards and referential.

In addition, the intervention time is important due to the lack of knowledge of the causes of the breakdown and the repair. During the diagnosis, the technician can identify a cause that can block the machine for days and sometimes weeks because of the repair or the availability of spare parts. There is then a huge and important loss of time from the assignment to the repair through the diagnosis and identification of the causes responsible for the breakdown.

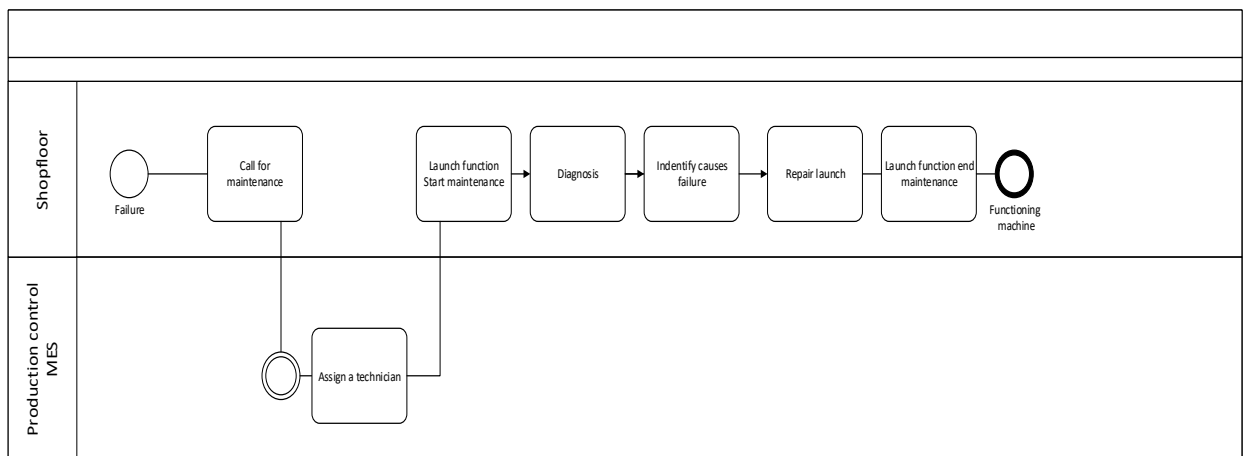


Fig. 2. Modeling of shopfloor maintenance.

4. Proposed Approach

In order to solve all the constraints that we have identified, we propose the diagram model represented in BPMN as shown in Fig. 3, a maintenance management model that allows to optimize and make reliable the maintenance intervention based on a reference base as well as an expert system established by the machine learning.

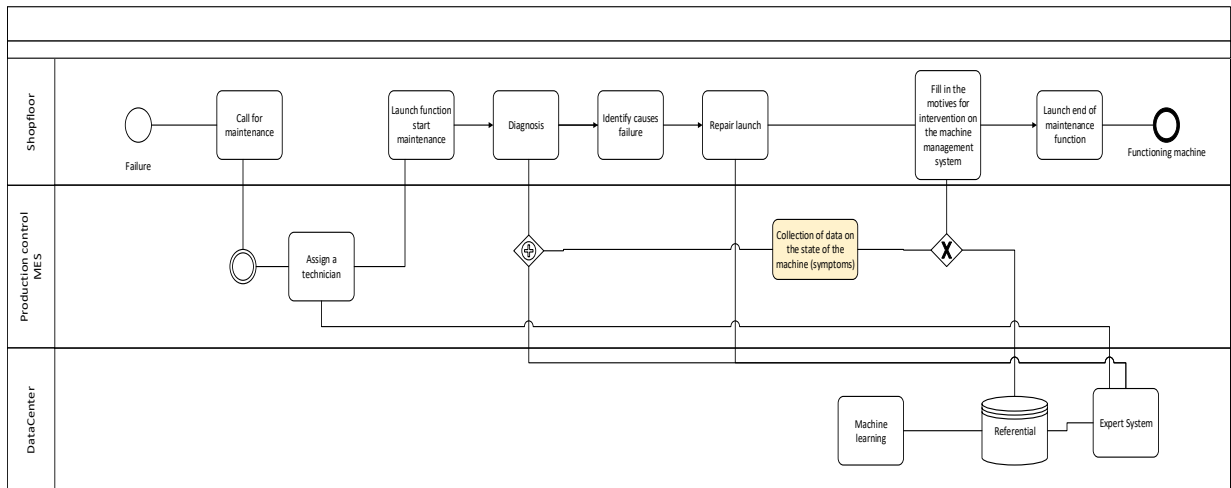


Fig. 3. Modeling of the maintenance process to integrate predictive maintenance.

Indeed, the BPMN can then be a beneficial approach in the Industry 4.0 domain, as well-defined processes will help to limit uncertainty in the decision-making process, eliminate redundant activities, and also manage resources in a rational way. In addition, the modularization of the process also facilitates communication and promotes transparency [11]. As a result, BPMN can be an asset and an interesting practice for all employees to understand a particular business process [10,12].

With the new expert system, the assignment of the technician will be done automatically following the proposal of the expert system according to the type and nature of the failure, this allows to reduce the time related to the assignment of the maintenance technician. The expert system will support the diagnosis and therefore it allows to quickly identify the location and the nature of the failure, so the data on the state of the machine during the diagnosis which are provided by the technician will be collected to enrich the reference base.

This expert system makes it possible to propose repair action plans for each failure mode and, as a result, for each failure the expert system will propose an action plan that includes the repair, the material to be changed, the time of the intervention and the repair operating mode, all the operations as well as the motives of intervention carried out by the maintenance technician at the time of the repair are collected to feed the reference base and subsequently develop and make the expert system more reliable.

To sum up, the expert system allows to drastically reduce the intervention time of the maintenance, and consequently the reduction of the machine downtime, and constitutes a basis for the implementation of predictive maintenance through machine learning and datamining. The machine learning includes algorithms that allow a computer system to learn by itself on the basis of previous knowledge and experience. Also, data mining consists in the application of machine learning methodologies to large databases [13].

A huge amount of data is produced in the maintenance of wire cutting machines as well as electrical test equipment, this data forms the basis for the implementation of predictive maintenance. In fact, the use of machine learning and data mining provides information and results that are useful tools to improve and update existing procedures and create new innovative techniques that reduce machine downtime as well as spare parts to maintain [1].

5. Conclusion

Maintenance activities in the aeronautical industry are considered more and more as a costly expense item, due to their impact on the company's performance. Therefore, to ensure the proper functioning of the machines, maintenance activities must be adapted and optimized. In this respect, predictive maintenance allows to reduce the unscheduled

downtime, in order to ensure the availability of the machines, by reducing the production stops. The challenge is obviously to better detect anomalies and anticipate breakdowns through the collection, processing and analysis of data, predictive maintenance is part of this logic. However, the transformation of current maintenance processes is essential in order to integrate artificial intelligence and apply a different maintenance than what was done in the past, and therefore to move from troubleshooting after a malfunction to a failure-free operation, in other words, a maintenance system able to propose very precise and reliable maintenance planning, specify which type of technical operations to do, which parts to change, and thus the time it takes to intervene, this planning will be taken into account in the production planning, and will allow the aeronautics industry to drastically reduce defects and costs, anticipate breakdowns and improve productivity. We plan in our future works to implement predictive maintenance by using the techniques of machine learning and data mining in order to make the maintenance process at the shop floor more reliable and optimized.

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